Lecture 10 - Product/Quotient Rule, Trig Derivatives

Garrett Credi

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Exam 1 Recap

1. All grades for Exam 1 released, regrade requests open until next Monday.
2. After Monday, curve will be determined
3. Recap: Essential Discontinuity, Vertical Tangent Singularity, Vertical Asymptotes

Also, Anonymous Feedback Form.

Quiz Time!

Let’s get into neat rows, spread out, and slay this quiz :)

Product and Quotient Rule

Long Term Goals

Recall we're trying to build a *synthetic* approach to studying derivatives:
Atomic Rules:

- Power Rule: $\frac{d}{dx}x^n$
- Trigonometric Rules: $\frac{d}{dx}\sin(x)$ etc.
- Exponential/Logarithm Rules: $\frac{d}{dx}e^x$, $\frac{d}{dx}\ln(x)$

Combination Rules:

- Sum/Difference Rule: $\frac{d}{dx}(f(x) \pm g(x))$
- Product Rule: $\frac{d}{dx}f(x)g(x)$
- Chain Rule: $\frac{d}{dx}g(f(x))$
- Constant Multiple Rule: $\frac{d}{dx}c \cdot f(x)$
- Quotient Rule: $\frac{d}{dx}\frac{f(x)}{g(x)}$
- Inverse Function Rule: $\frac{d}{dx}f^{-1}(x)$

Product Rule I

We want to determine what $\frac{d}{dx}(f(x) \cdot g(x))$ is, hopefully in terms of simpler derivatives involving $f(x), g(x)$.
Is it just $f'(x)g'(x)$? NO NO NO NO NO!



Figure: My one wish willow is for you to not compute derivates of products like this!

Product Rule II

Well, then what is it?
Approach 1: Linearization! We motivated the other rules by examining what the operation did to linear functions first - let's try that here!
Linearization of $f(x)$ at $x = x_0$: $f(x_0) + f'(x_0)(x - x_0)$,
Linearization of $g(x)$ at $x = x_0$: $g(x_0) + g'(x_0)(x - x_0)$,
Linearization of $f(x)g(x)$ at $x = x_0$:

$$f(x)g(x) \approx (f(x_0) + f'(x_0)(x - x_0)) \cdot (g(x_0) + g'(x_0)(x - x_0)) \quad (1)$$
$$= f(x_0)g(x_0) + f'(x_0)g(x_0)(x - x_0) + f(x_0)g'(x_0)(x - x_0) + f'(x_0)g'(x_0)(x - x_0)^2 \quad (2)$$
$$= f(x_0)g(x_0) + (f'(x_0)g(x_0) + f(x_0)g'(x_0))(x - x_0) + f'(x_0)g'(x_0)(x - x_0)^2 \quad (3)$$

So why is it NOT $f'(x)g'(x)$? Because that's the coefficient of something involving a *quadratic* term, not linear!
The derivative would be the slope on the *linear* term.

Product Rule III

The Product Rule

$$\frac{d}{dx}(f(x) \cdot g(x)) = f'(x)g(x) + f(x)g'(x) \tag{4}$$

Product Rule Examples I

Example 1: What is the derivative of:

$$x^2 e^x + \frac{2^x}{x}? \tag{5}$$

We compute:

$$\frac{d}{dx}(x^2 e^x + \frac{2^x}{x}) = \frac{d}{dx}(x^2 e^x) + \frac{d}{dx}(\frac{2^x}{x}) \tag{6}$$

$$= (\frac{d}{dx}x^2) \cdot (e^x) + (x^2) \cdot (\frac{d}{dx}e^x) + \frac{d}{dx}(2^x \cdot \frac{1}{x}) \tag{7}$$

$$= (2x) \cdot e^x + x^2 \cdot e^x + (\frac{d}{dx}2^x) \cdot (\frac{1}{x}) + (2^x) \cdot (\frac{d}{dx}\frac{1}{x}) \tag{8}$$

$$= 2xe^x + x^2 e^x + \ln(2)2^x \cdot \frac{1}{x} + 2^x \cdot \frac{-1}{x^2} \tag{9}$$

Product Rule Examples II

For a more applied example: Let's suppose you drive $m(t)$ miles per day, in a car whose fuel efficiency is given by $e(t)$ gallons/mile where the price of fuel is given by $p(t)$ dollars/gallon, t is measured in days.
Suppose we've measured the relevant quantities at:

$$m(1) = 50mi, \quad e(1) = \frac{1}{25}\frac{gal}{mi}, \quad p(1) = 4.15\frac{\$}{gal} \tag{10}$$

$$m'(1) = 0\frac{mi}{day}, \quad e'(1) = \frac{1}{1000}\frac{\frac{gal}{mi}}{day}, \quad p'(1) = 0.01\frac{\frac{\$}{gal}}{day} \tag{11}$$

What is the rate of change of the cost to drive your car on day 1?

Product Rule Examples III

Step 1: Cost Function. $C(t) = m(t) \cdot e(t) \cdot p(t)$.
Step 2: Derivative.

$$C'(t) = \frac{d}{dt}(m(t) \cdot e(t) \cdot p(t)) = \frac{d}{dt}((m(t)e(t)) \cdot p(t)) \tag{12}$$

$$= \frac{d}{dt}(m(t) \cdot e(t)) \cdot p(t) + (m(t) \cdot e(t)) \cdot p'(t) \tag{13}$$

$$= m'(t)e(t)p(t) + m(t)e'(t)p(t) + m(t)e(t)p'(t) \tag{14}$$

Product Rule Examples IV

Step 3: Compute.

$$C'(1) = m'(1)e(1)p(1) + m(1)e'(1)p(1) + m(1)e(1)p'(1) \tag{15}$$

$$= 0 \cdot \frac{1}{25}\frac{gal}{mi} \cdot 4.15\frac{\$}{gal} \tag{16}$$

$$+ 50mi \cdot \frac{1}{1000}\frac{\frac{gal}{mi}}{day} \cdot 4.15\frac{\$}{gal} \tag{17}$$

$$+ 50mi \cdot \frac{1}{25}\frac{gal}{mi} \cdot 0.01\frac{\frac{\$}{gal}}{day} \tag{18}$$

$$= 0\frac{\$}{day} + 0.2075\frac{\$}{day} + 0.02\frac{\$}{day} \tag{19}$$

$$= 0.2275\frac{\$}{day} \tag{20}$$

Quotient Rule I

Similarly, we'd want to determine $\frac{d}{dx}\frac{f(x)}{g(x)}$, again in terms of simpler derivatives involving $f(x)$, $g(x)$.

Is it just $\frac{f'(x)}{g'(x)}$? SUPER NO! NO! NO!



Figure: I'm amending my one wish willow to also include you not computing derivatives of quotients like this.

Quotient Rule II

We can determine what it actually is by realizing that:

$$\frac{f(x)}{g(x)} = f(x) \cdot \frac{1}{g(x)} \tag{21}$$

and since we know how to linearize $f(x)$ and how to linearize a product, we need only understand how to linearize something like $\frac{1}{g(x)}$.

First, we note that for the derivative of $\frac{f(x)}{g(x)}$ at $x = x_0$ to exist, the function $\frac{f(x)}{g(x)}$ must be continuous at $x = x_0$, so $g(x_0) \neq 0$.

This allows us to approximate $\frac{1}{g(x)} \approx \frac{1}{g(x_0) + g'(x_0)(x - x_0)}$.

Quotient Rule III

Since

$$\frac{1}{g(x)} \approx \frac{1}{g(x_0) + g'(x_0)(x - x_0)} = \frac{1}{(g(x_0) - g'(x_0)x_0) + g'(x_0)x} \tag{22}$$

we can focus on understanding how to linearize functions like

$$\frac{1}{a + bx}. \tag{23}$$

We compute:

$$\frac{d}{dx}\frac{1}{a+bx} = \frac{d}{dx}\frac{1}{b}\frac{1}{x + \frac{a}{b}} = \frac{1}{b}\frac{d}{dx}\frac{1}{x + \frac{a}{b}} \tag{24}$$

$$= \frac{1}{b}\frac{-1}{(x + \frac{a}{b})^2} = \frac{1}{b}\frac{-1}{x^2 + \frac{2a}{b}x + \frac{a^2}{b^2}} \tag{25}$$

$$= \frac{-1}{bx^2 + 2ax + \frac{a^2}{b}} = \frac{-b}{b^2x^2 + 2abx + a^2} = \frac{-b}{(a+bx)^2} \tag{26}$$

Quotient Rule IV

And so,

$$\frac{1}{g(x)} \approx \frac{1}{g(x_0)} + \frac{-g'(x_0)}{g(x_0)^2}(x - x_0) \tag{27}$$

meaning that:

$$\frac{f(x)}{g(x)} = f(x) \cdot \frac{1}{g(x)} \tag{28}$$

$$= (f(x_0) + f'(x_0)(x - x_0)) \cdot (\frac{1}{g(x_0)} + \frac{-g'(x_0)}{g(x_0)^2}(x - x_0)) \tag{29}$$

$$= \frac{f(x_0)}{g(x_0)} + \frac{f'(x_0)}{g(x_0)}(x - x_0) + \frac{-f(x_0)g'(x_0)}{g(x_0)^2}(x - x_0) + O((x - x_0)^2) \tag{30}$$

$$= \frac{f(x_0)}{g(x_0)} + \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}(x - x_0) \tag{31}$$

Quotient Rule V

Moral of the story: it's HARD to actually get the derivative of a quotient so in no universe would the formula be as easy as $\frac{f'(x)}{g'(x)}$. It is actually:

Quotient Rule

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2} \tag{32}$$

Quotient Rule Example I

E.g. computing:

$$\frac{d}{dx}\frac{2xe^x + 3x^2}{x^3 + 2x + 1} \tag{33}$$

$$\frac{d}{dx}\frac{2xe^x + 3x^2}{x^3 + 2x + 1} = \frac{(\frac{d}{dx}(2xe^x + 3x^2)) \cdot (x^3 + 2x + 1) - (2xe^x + 3x^2) \cdot (\frac{d}{dx}(x^3 + 2x + 1))}{(x^3 + 2x + 1)^2} \tag{34}$$

$$= \frac{(\frac{d}{dx}(2xe^x) + \frac{d}{dx}(3x^2)) \cdot (x^3 + 2x + 1) - (2xe^x + 3x^2) \cdot (3x^2 + 2)}{(x^3 + 2x + 1)^2} \tag{35}$$

$$= \frac{(2e^x + 2xe^x + 6x) \cdot (x^3 + 2x + 1) - (2xe^x + 3x^2) \cdot (3x^2 + 2)}{(x^3 + 2x + 1)^2} \tag{36}$$

Trigonometric Derivatives

Derivative of $\sin(x)$ I

Recall from our discussion on the Squeeze Theorem:

$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1 \tag{37}$$

$$\lim_{x \to 0} \frac{\cos(x) - 1}{x} = 0 \tag{38}$$

What is this limit in terms of derivatives?

$$\frac{d}{dx} |_{x=0} \sin(x) = 1, \frac{d}{dx} |_{x=0} \cos(x) = 0 \tag{39}$$

Derivative of $\sin(x)$ II

What about the derivative function?

$$\frac{d}{dx}\sin(x) = \lim_{h\to 0}\frac{\sin(x+h) - sin(x)}{h} \tag{40}$$

$$= \lim_{h\to 0}\frac{\sin(x)\cos(h) + \sin(h)\cos(x) - \sin(x)}{h} \tag{41}$$

$$= \lim_{h\to 0}\frac{\sin(x)(\cos(h) - 1) + \cos(x)\sin(h)}{h} \tag{42}$$

$$= \lim_{h\to 0}\left(\sin(x)\frac{\cos(h) - 1}{h} + \cos(x)\frac{\sin(h)}{h}\right) \tag{43}$$

$$= \sin(x)\lim_{h\to 0}\left(\frac{\cos(h) - 1}{h}\right) + \cos(x)\lim_{h\to 0}\left(\frac{\sin(h)}{h}\right) \tag{44}$$

$$= \sin(x)\cdot 0 + \cos(x)\cdot 1 = \cos(x) \tag{45}$$

Derivative of $\sin(x)$ III

Derivative of $\sin(x)$

$$\frac{d}{dx}\sin(x) = \cos(x) \tag{46}$$

What about for $\cos(x)$?

Derivative of $\cos(x)$ I

$$\frac{d}{dx}\cos(x) = \lim_{h\to 0}\frac{\cos(x+h)-\cos(x)}{h} \tag{47}$$

$$= \lim_{h\to 0}\frac{\cos(x)\cos(h)-\sin(x)\sin(h)-\cos(x)}{h} \tag{48}$$

$$= \cos(x)\lim_{h\to 0}\left(\frac{\cos(h)-1}{h}\right) - \sin(x)\lim_{h\to 0}\left(\frac{\sin(h)}{h}\right) = -\sin(x) \tag{49}$$

Derivative of $\cos(x)$

$$\frac{d}{dx}\cos(x) = -\sin(x) \tag{50}$$

Derivative of $\tan(x)$

What about for $\tan(x)$? Do we need to do even more limit yuckiness?

No! Since $\tan(x) = \dfrac{\sin(x)}{\cos(x)}$ we can just use quotient rule!

$$\frac{d}{dx}\tan(x) = \frac{d}{dx}\frac{\sin(x)}{\cos(x)} \tag{51}$$

$$= \frac{\left(\frac{d}{dx}\sin(x)\right)\cdot\cos(x) - \sin(x)\cdot\left(\frac{d}{dx}\cos(x)\right)}{\cos(x)^2} \tag{52}$$

$$= \frac{\cos(x)\cdot\cos(x) - \sin(x)\cdot(-\sin(x))}{\cos(x)^2} \tag{53}$$

$$= \frac{\cos(x)^2 + \sin(x)^2}{\cos(x)^2} \tag{54}$$

$$= \frac{1}{\cos(x)^2} = \sec(x)^2 \tag{55}$$

Derivative of $\sin(2x)$

While we'll get an easier method to do this next lecture, we can also use trig rules to compute $\frac{d}{dx}\sin(2x)$:

$$\frac{d}{dx}\sin(2x) = \frac{d}{dx}(2\sin(x)\cos(x)) \tag{56}$$

$$= 2\frac{d}{dx}(\sin(x)\cos(x)) \tag{57}$$

$$= 2 \cdot \left(\left(\frac{d}{dx}\sin(x)\right) \cdot \cos(x) + \sin(x) \cdot \left(\frac{d}{dx}\cos(x)\right)\right) \tag{58}$$

$$= 2 \cdot (\cos(x) \cdot \cos(x) + \sin(x) \cdot (-\sin(x))) \tag{59}$$

$$= 2 \cdot (\cos(x)^2 - \sin(x)^2) \tag{60}$$

$$= 2 \cdot \cos(2x) \tag{61}$$

Worksheet Time!

Let's get into groups (or if you really like working alone, please stay so I can be here to help :)) and work on the worksheet!

End of Class!

That’s all folks!